

MAGNET RUNDOWN UNIT
GE MODEL 46-294231G1
MANUAL PART NO. 46-318393
AMI MODEL GE-MRU
SERVICE MANUAL
REVISION 10
07 MARCH 2001

CONTENTS

1.0 PREINSTALLATION

2.0 INSTALLATION

3.0 FUNCTIONAL TEST

4.0 THEORY OF OPERATION

5.0 SERVICE ADJUSTMENTS

6.0 SCHEMATICS, FRAME ASSEMBLY, WIRING HARNESS

7.0 REPLACEMENT PARTS LIST

8.0 PERIODIC MAINTENANCE

1.0 PREINSTALLATION

- 1.01 Each MRU is shipped with an operating manual and service manual. The batteries are installed and were fully charged before leaving the factory. The charge on the batteries should be verified at 20 volts (minimum) before any adjustments are made.
- 1.02 Verify that AC power of 115V/230V (nominal), 50-60Hz, 35 watts (maximum) is available within 6 ft. of the mounting location. MRU's are shipped strapped for 115V.
- 1.03 Inspect mounting surface and obtain the type of mounting fasteners required for the surface involved. Use fasteners capable of supporting a 32 lb. (14.5kg) load. The case is drilled for 1/4" or 6 mm bolts arranged in a rectangular pattern. Vertical spacing is 5.50 inches and horizontal spacing is 10.00 inches. Use flat washers under the fasteners.
- 1.04 ALL 200-240 volt applications MUST be PERMANENTLY INSTALLED. 100-120 volt applications may be PERMANENTLY INSTALLED if desired. For any PERMANENTLY INSTALLED application, refer to the SERVICE MANUAL schematic for connections to TB1. All PERMANENT installations shall be connected to a dedicated Mains circuit, with a disconnect device that is in compliance with local regulatory standards.

If conduit is to be used, contact building maintenance or an outside contractor as required to have conduit/wiremold installed.

- 1.05 Verify that magnetic field intensity is less than 200 Gauss at the mounting location. See 1.5T manual for details.
- 1.06 A No. 2 Phillips screwdriver is required to open the front cover.

2.0 INSTALLATION AND ADJUSTMENT

- 2.01 Mount the MRU to the wall surface. Connect AC power.
- 2.02 Connect a DVM, Simpson Model 470 or equivalent, positive to the test point labeled +20.7V and the negative to the test point marked ground. The multimeter should read 20.7 VDC. Adjust R3 if necessary to obtain 20.7 V. Verify CHARGER POWER LED on the front panel is lit.
- 2.03 ***WARNING: ENSURE THE HEATERS ARE NOT CONNECTED TO J2 TO PREVENT INADVERTENT MAGNET QUENCH.***

- 2.04 Remove the 20 ohm 5% ceramic resistor from the plastic bag and connect it between lugs 1 and 2 of TB2.

WARNING: *THE 20 OHM TEST RESISTOR WILL BECOME HOT DURING THIS TEST.*

- 2.05 Install fuse F2 (2.5A T 5x20mm fuse). Measure the voltage at test point +20.7V with respect to ground; it should be between 20.1 and 20.7 volts. If the MRU batteries have been stored for more than one year without charging, it may be necessary to wait for up to 8 hours for the voltage to reach 20.7 volts.
- 2.06 Depress and release the RUN DOWN button. Measure the elapsed time until the RUN DOWN push-button pops back out. The time must be a minimum of 30 seconds. The red HEATER ACTUATED LED will light and remain lit until reset with switch S6. Press push-button switch S6 on the printed circuit board to reset the HEATER ACTUATED LED.
- 2.07 Remove the test resistor from TB2. Disconnect the DVM and close the front cover.
- 2.08 Verify the green CHARGER POWER LED is lit.
- 2.09 Depress and hold the TEST BATTERY switch for 15 seconds and the green BATTERY TEST LED should light. The set point is 16.5 volts.

WARNING: *BEFORE CONNECTING P2 TO J2, ASSURE THAT AC POWER HAS BEEN APPLIED TO THE MRU FOR AT LEAST 3 MINUTES AND THAT THE RUN DOWN BUTTON HAS NOT BEEN ACTUATED FOR AT LEAST 3 MINUTES. FAILURE TO HEED THIS WARNING MAY CAUSE A MAGNET QUENCH.*

- 2.10 Open the cover and connect the main coil heater cable plug P2 to matching connector J2. Close and fasten the cover. Depress the TEST HEATER switch. The green HEATER TEST LED should light.
- 2.11 Complete the maintenance log with date and signature.

3.0 FUNCTIONAL TEST

- 3.01 Verify the green CHARGER POWER LED is lit.
- 3.02 Depress and hold the TEST BATTERY switch for 15 seconds. The green BATTERY TEST LED should light.
- 3.03 Depress the TEST HEATER switch. The green HEATER TEST LED should light.

4.0 THEORY OF OPERATION

4.01 General System Description

The Model 46-294231G1 MRU is designed to provide a timed (30seconds minimum, 50 seconds nominal) 800 mA DC current source to both of the heaters in a 1.5T magnet system. In addition, test circuits are provided to verify battery charger operation, battery charge condition and heater resistance limits.

4.02 AC power supply and regulator

115 VAC is permanently connected through fuse F1 to T1. A full wave rectifier CR4 and C1 provide filtered DC to U1, an LM350T IC regulator. U1 has internal short circuit and over-temperature limiting. Diodes CR2 and CR1 provide transient and reverse polarity protection. R3 is used to set the float charge on the battery assembly to 20.7 volts. DS1 provides a continuous indication of regulator output with isolation provided by diode CR3 to prevent lamp indication with AC input or regulator failure. During the BATTERY TEST function, a 15 ohm test load is placed across the battery and the regulator is disabled by the second half of S1 to ensure that the battery is the only source of power.

4.03 Test Circuits

During the BATTERY TEST mode S1 also applies an input via R27 to a comparator circuit consisting of 1/4 of U4 and associated components. U5 provides a constant current source to a voltage divider to establish a 16.5 volt trip point reference. When R26 is properly set, LED DS2 (BATTERY TEST) will light at 16.5 volts and above and not light below 16.5 volts. Resistors R29, R30 and capacitor C7 provide a low pass filter and transient protection to the comparator input.

Two other sections of U4 are used to sense upper and lower limit heater resistance during the HEATER TEST mode. Pressing S2 to either the A or B position allows approximately 5mA to flow from the battery through R5 and the heater element under test.

The resulting voltage drop across R5 is sensed via R16 at two comparators. R8 sets a 12 ohm lower limit and R20 a 32 ohm upper limit for heater resistance. If the heater resistance is between 12 ohms and 32 ohms, the HEATER TEST LED will light. Since this is a critical test, a switch is provided to TEST HEATER LED, S5. U2 and U3 provide constant current sources for stable reference voltage dividers. R12 through R15, C4 and C5 provide low pass filtering and transient protection.

4.04 Time Delay and Output Circuits

Actuation of the covered push-button RUN DOWN switch S4, energizes both heaters by resetting a thermally timed circuit breaker S4. The 100mA rated sensing element passes approximately 170 mA for a nominal 50 seconds at which time the circuit breaker opens, interrupting current flow through the heaters. The power supplied to the circuit breaker heating element is limited by resistor R34. The red LED marked HEATER ACTUATED will light and stay lit whenever the RUN DOWN switch is activated, or if the HEATER TEST switch is activated and the heater circuits are open. The latching circuit should be reset only by General Electric Field Service personnel. S6, a momentary push-button switch is located on the main PC board and resets the SCR latching circuit.

ACCIDENTAL HEATER ACTIVATION WILL RESULT IN LOSS OF EXPENSIVE COOLING FLUIDS AND MAY CAUSE PERMANENT DAMAGE TO THE 1.5T MAGNET.

5.0 SERVICE ADJUSTMENTS

WARNING: *DO NOT USE A VOM FOR SERVICE. SUFFICIENT VOLTAGE AND/OR CURRENT IS PRESENT IN MANY VOMS TO CAUSE CIRCUIT DAMAGE. 3-1/2 DIGIT ACCURACY IS REQUIRED FOR SOME ADJUSTMENTS. USE A DVM SUCH AS A SIMPSON MODEL 470 OR EQUIVALENT.*

No connections or subassemblies need to be removed for service adjustments except as noted.

For specific details of service adjustments see Section 8.0, Periodic Maintenance.



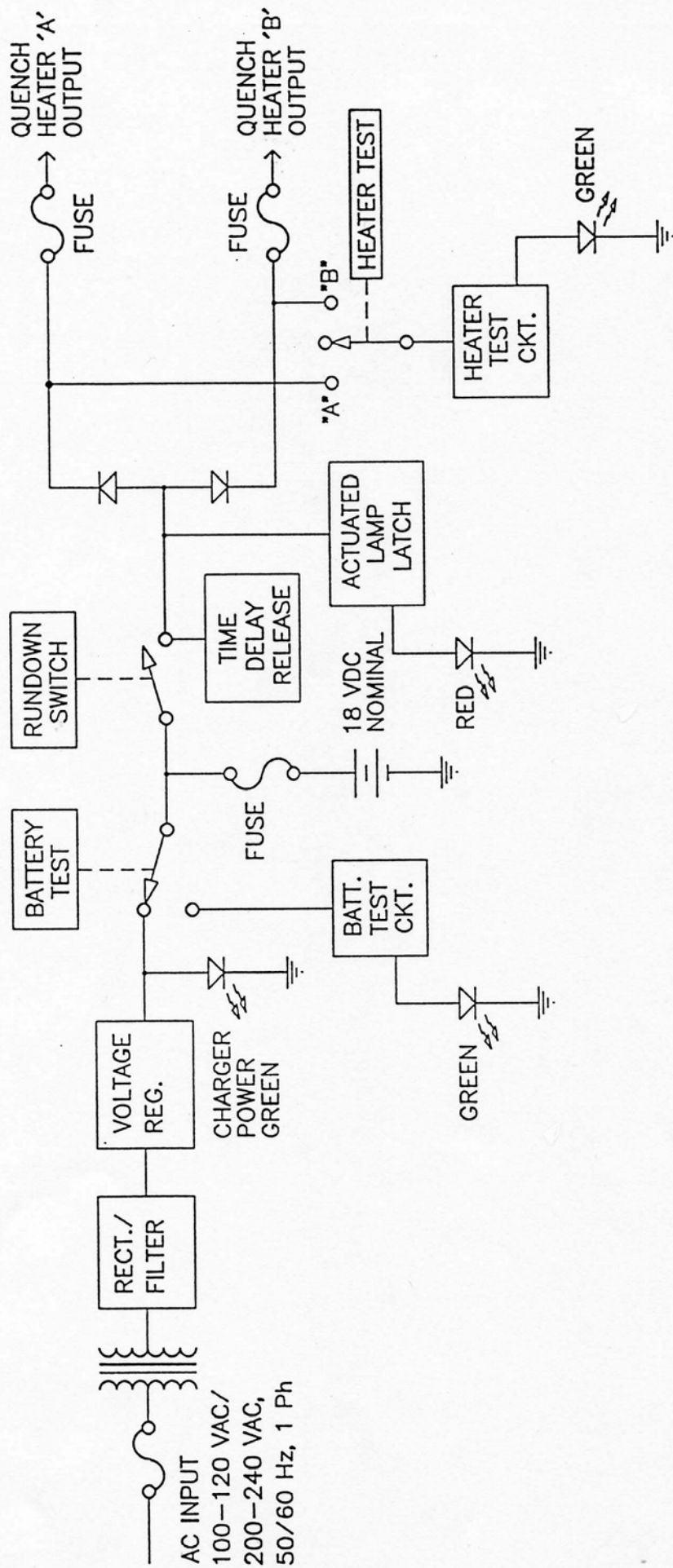


FIGURE 2
MAGNET RUNDOWN UNIT FUNCTIONAL BLOCK DIAGRAM

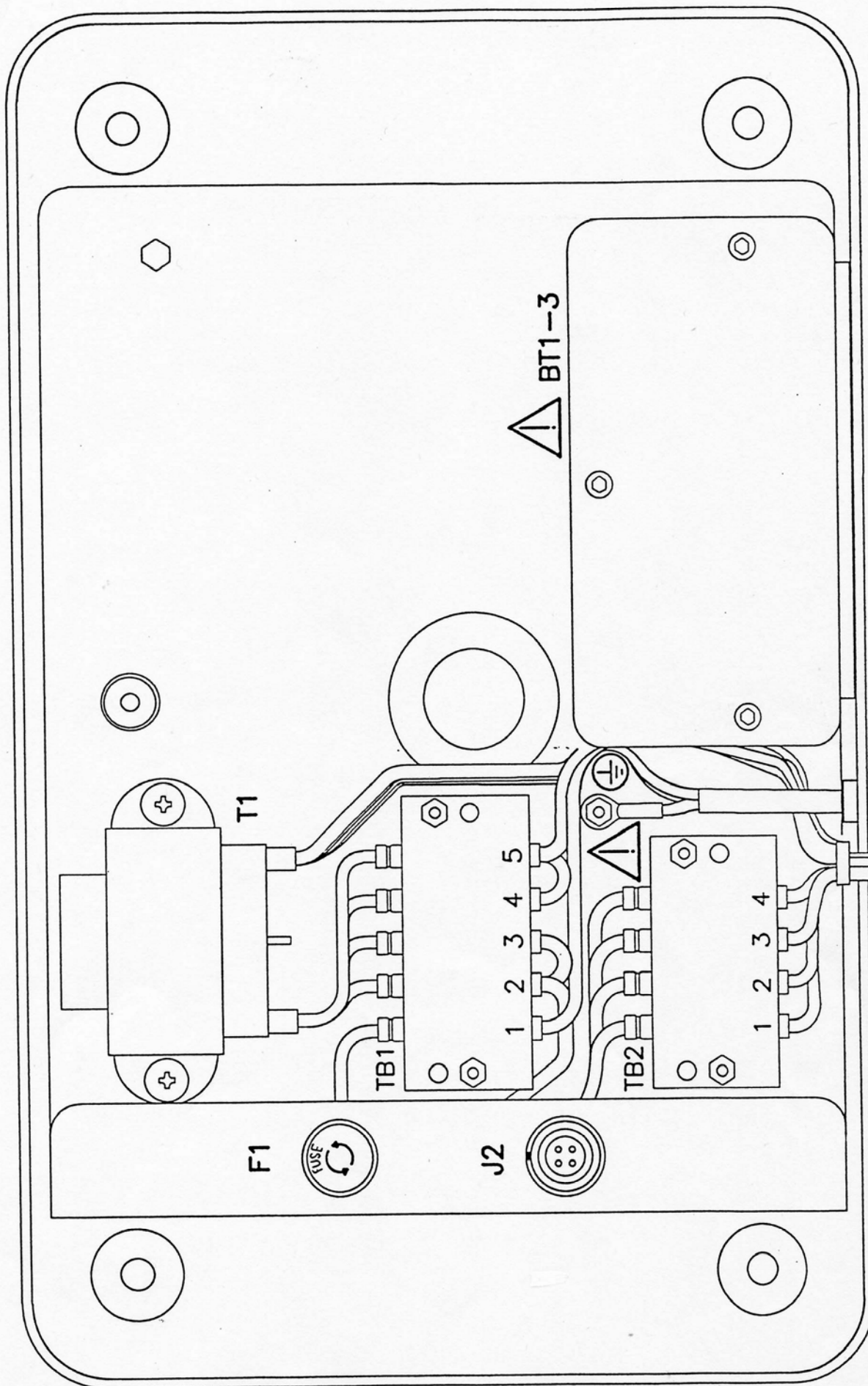


FIGURE 3
MAGNET RUNDOWN UNIT MECHANICAL LAYOUT

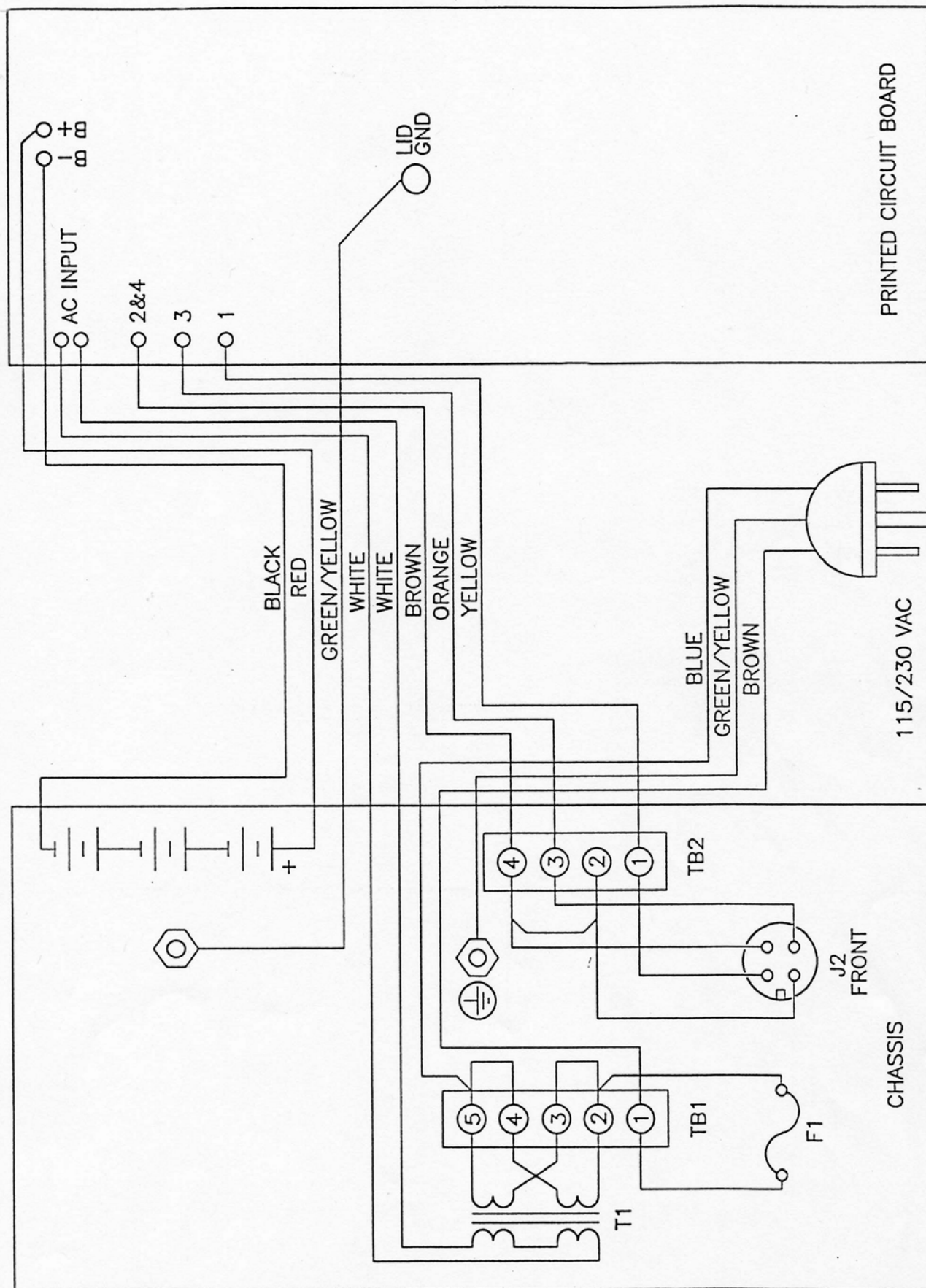
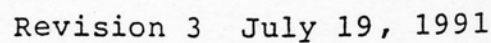
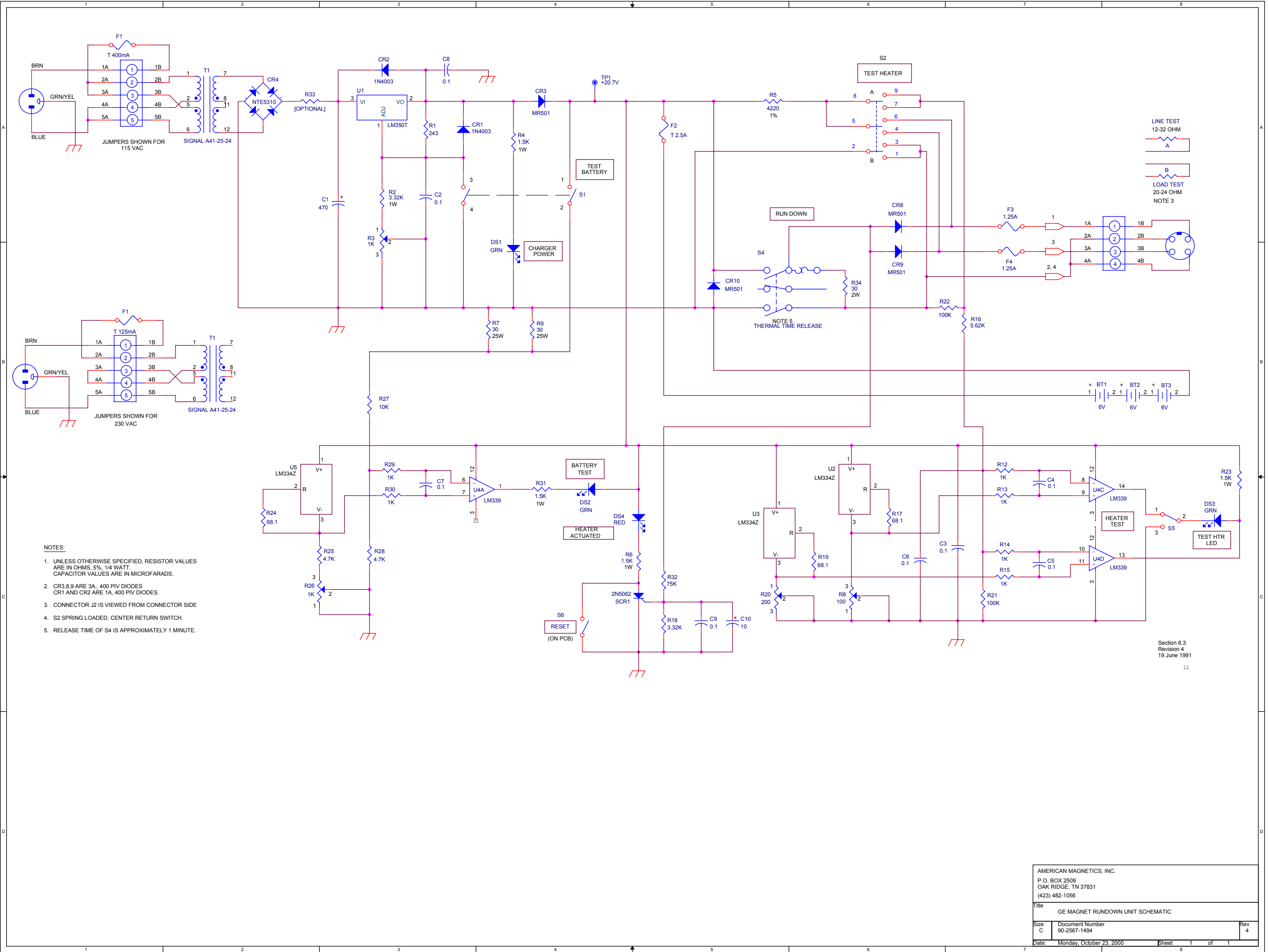


FIGURE 4
MAGNET RUNDOWN UNIT WIRING DIAGRAM





- NOTES:
1. UNLESS OTHERWISE SPECIFIED, RESISTOR VALUES ARE IN OHMS, 5%, 1/4 WATT. CAPACITOR VALUES ARE IN MICROFARADS.
 2. CR3,8,9 ARE 3A., 400 PIV DIODES. CR1 AND CR2 ARE 1A., 400 PIV DIODES.
 3. CONNECTOR J2 IS VIEWED FROM CONNECTOR SIDE.
 4. S2 SPRING LOADED, CENTER RETURN SWITCH.
 5. RELEASE TIME OF S4 IS APPROXIMATELY 1 MINUTE.

Section 6.3
Revision 4
19 June 1991

7.0 REPLACEMENT PARTS LIST

*BT1-3.....	3	Battery, Gel Cel 6V 1.2AH Yuasa NP1.2-6.....	GS-4063
*C1	1	Capacitor, 470 uF 63V electrolytic Sprague 517D477M063DKGA.....	EC-1140
C2-9	8	Capacitor, 0.1uF 50V ceramic Sprague 1C20Z5U104M050B.....	EC-1145
C10	1	Capacitor, 10 uF 25V tantalum	EC-1120
CR1,2.....	2	Diode, 1N4003	ED-1056
CR3,8-10	4	Diode, Motorola MR501	ED-1060
CR4.....	1	Bridge rectifier NTE 5310.....	EI-1183
DS1-3.....	3	LED, green	EL-1015
DS4	1	LED, red.....	EL-1000
*F1.....	2	Fuse, 0.400A Bussman GDC Time-Lag (100-120V)or Littelfuse 239.400	EF-1001
		Fuse, 0.125A Bussman GDC Time-Lag (200-240V).....	EF-1002
*F2.....	1	Fuse, 2.5A Bussman GDC Time-Lag.....	EF-1008
*F3,4.....	4	Fuse, 1.25A Bussman GDB	EF-1004
J2	1	Connector, Lemo EGG2B304CNL	EH-1815
R1	1	Resistor, 243 ohm, RN55D	ER-1023
R2	1	Resistor, 3.3K ohm, 1W, IRC RL32	ER-1038
R3,26	2	Potentiometer, 1K ohm, Bourns 3006P-1-102	EP-1111
R4,6,23,31	4	Resistor, 1.5K ohm, RN65C.....	ER-1033
R5	1	Resistor, 4.22K ohm, RN55D	ER-1043
R7,9	2	Resistor, 30 ohm 25W 1% Dale RH25-30	ER-2012
R8	1	Potentiometer, 100 ohm, Bourns 3006P-1-101	EP-1007
R12-15,29,30	6	Resistor, 100K ohm, RN55D	ER-1030
R16	1	Resistor, 5.62K ohm, RN55D	ER-1048
R17,19,24	3	Resistor, 68.1 ohm, RN55D	ER-1008
R18	1	Resistor, 3.32K ohm RN55D	ER-1037
R20	1	Potentiometer, 200 ohm, Bourns 3006P-1-201	EP-1008
R21,22	2	Resistor, 100K ohm, RN55D	ER-1125
R25,28	2	Resistor, 4.7K ohm, 1/4W, 10%.....	ER-2095
R27	1	Resistor, 10K ohm, RN55D	ER-1065
R32	1	Resistor, 75K ohm, RN55D	ER-1115
R34	1	Resistor, 30 ohm, 2W, 5%.....	ER-2014
S1,5.....	2	Switch, DPDT mom push Alco MPA-206R-PC	ES-1095
S2.....	1	Switch, 3PDT center-off spring-return C&K 7305SYW40	ES-1018
S4.....	1	Circuit breaker, 0.1A, E-T-A 46-400-L10-A3-SI.....	EF-1100
S6.....	1	Switch, SPST mom push Alco FSM-2	ES-1070
SCR1	1	Silicon controlled rectifier, 2N5062.....	ET-1032
T1.....	1	Transformer, Signal A41-25-24	EX-1040
TB1.....	1	Terminal block, Cinch 4-164.....	EH-3008
TB2.....	1	Terminal block, Cinch 6-140.....	EH-3012
U1	1	Voltage regulator, LM350T	EI-1028
U2,3,5	3	Constant current source, LM334	EI-1023
U4	1	Quad comparator, LM339N	EI-1132
	4	LED mounting hardware.....	EL-1005
	1	Printed circuit board 90-2567-1494R4 Rev1.....	EB-1100
	1	Fuseholder, Littelfuse 345623.....	EF-1510
	14	Fuseclip, Littelfuse 111501	EF-1515
	2	Button, green, Alco C23-5	ES-1097

*Field replaceable part

1	Button, red, NKK AT412C	ES-1048
2	Guard, Alco G13	ES-1099
1	Heatsink, Wakefield 290-1AB	EI-1148
1	Cord, line, Belden 17513C-10	EH-2323
2	Marker strip, Cinch MS-4-140	FM-8006
2	Marker strip, Cinch MS-5-140	FM-7090
1	Case assembly, Dwg. No. 90-2567-1482R3	FM-9000
1	Mounting plate, Dwg. No. 90-2567-1483R3	FM-9005
1	Battery cover, Dwg. No. 90-2567-1483R3	FM-9006
2	Plug, hole, 7/8"	EH-3103
1	Strain relief, International Components PGME-9F	EH-6075
1	Locking terminal lug, uninsulated	EH-4045
1	Standoff, 8-32x5-1/4", F-F	EH-2860
3	Standoff, 6-32x3", F-F	EH-2870
4	Screw, 4-40x1/4", steel, pan head	EH-4065
2	Screw, 4-40x3/8", steel, pan head	EH-4055
2	Screw, 4-40x1/2", steel, cap screw	GH-1156
4	Screw, 6-32x1/2", steel, pan head	EH-4060
4	Screw, 6-32x3/4", steel, truss head	EH-4085
1	Screw, 8-32x1/2", steel, flat head	EH-5060
6	Screw, 8-32x5/16", steel, hex head	GH-1202
1	Screw, 8-32x3/4", steel, truss head, Phillips	EH-5065
2	Screw, 10-32x3/8", Phillips, pan head	EH-5000
4	Nut, 4-40, steel	EH-4030
16	Nut, 6-32, steel	EH-4040
2	Nut, 10-32, steel	GH-1039
8	Lockwasher, #4	EH-4000
8	Lockwasher, #6	EH-4006
4	Lockwasher, #8	GH-1016
2	Lockwasher, #10	GH-1360
1	Flatwasher, #8	EH-3690
25	Ring terminals, #6	EH-3030
1	Locking ground lug, BNC	EH-4016
1	Nut, steel, 6-32, green	EH-4037
1	Screw, 6-32x3/8" steel, pan head, green	EH-4036
6	Faston connector, insulated, Tyton, PV22EQDF	EH-3040
1	Standoff, aluminum, 3/8x5.5", 8-32	EH-2862

8.0 PERIODIC MAINTENANCE

8.1 Weekly User Testing

- 8.11 Verify green CHARGER POWER LED is lit.
- 8.12 Depress the TEST BATTERY switch; green BATTERY TEST LED should light. If green BATTERY TEST LED does *not* light, then replace the batteries and perform yearly service tests (see sections 8.21 through 8.26).
- 8.13 Place the TEST HEATER toggle switch in the A position, green HEATER TEST LED should light.
- 8.14 Place the TEST HEATER toggle switch in the B position, green HEATER TEST LED should light.

8.2 Yearly, General Electric Field Service

- 8.20 Perform tests 8.11 through 8.14.
- 8.21 Remove heater connector P2 from J2.

WARNING: *ENSURE THE HEATERS ARE NOT CONNECTED TO J2 TO PREVENT INADVERTENT MAGNET QUENCH.*

- 8.22 Open the front cover and visually inspect the entire unit for any evidence of overheating, discoloration, etc. Remove fuse F2, 2.5A, remove batteries and hold down plate; visually inspect battery terminals for any evidence of corrosion. If any corrosion is evident, replace batteries and/or battery connecting lugs. If new batteries or lugs are needed, apply a thin layer of silicone grease or petroleum jelly to terminals after assembly.
- 8.23 Reinstall battery hold down plate and fuse F2, 2.5A. Locate 20 ohm 5% 20 watt ceramic resistor in plastic bag, remove and connect to lugs 1 and 2 of TB2. Connect the DVM across this resistor and press the RUN DOWN push-button. The DVM should read approximately 20 volts for approximately 50 seconds. The test resistor must remain energized for a minimum of 30 seconds. Allow a minimum of 3 minutes for switch S4 to cool before subsequent test resistor timing cycles.
- 8.24 Move the test resistor from lugs 1 and 2 of TB2 to lugs 3 and 4 of TB2. Connect the DVM across this resistor and press the RUN DOWN push-button. The DVM should read approximately 20 volts for approximately 50 seconds. The test resistor must remain on for a minimum of 30 seconds. Remove the test resistor from TB2.

- 8.25 Connect the DVM to TP +20.7V and TP ground. The meter should read 20.7V +/- 0.1V. If not, remove fuse F2 and adjust R3 for 20.7V and then replace F2.

WARNING: BEFORE CONNECTING P2 TO J2, ASSURE THAT AC POWER HAS BEEN APPLIED TO THE MRU FOR AT LEAST 3 MINUTES AND THAT THE RUN DOWN BUTTON HAS NOT BEEN ACTUATED FOR AT LEAST 3 MINUTES. FAILURE TO HEED THIS WARNING MAY CAUSE A MAGNET QUENCH.

- 8.26 Disconnect the DVM, plug P2 into J2, close the cover, perform tests 8.11 through 8.14. Log the test date and time.

8.3 Every 3 Years

- 8.31 Perform the same tests as yearly, Section 8.2 except replace all three 6V 1.2AH Gel Cells (PN GS-4063). If not in inventory, these can normally be obtained from any electronics supply house or burglar alarm company.

OBSERVE POLARITY.

- 8.32 If the Gel Cells have been stored for one year or more, it may be necessary to wait for as long as 8 hours for battery voltage to reach 20.0 volts minimum. Initial charge current is approximately 800 mA which will decrease to approximately 3 mA when fully charged.

8.4 Every 6 Years

- 8.41 Disconnect AC power, remove fuse F2, 2-1/2 amp, and disconnect heater plug P2 from J2. Remove the main PC board and unsolder C1, 470 uF 63 volt electrolytic capacitor (PN EC-1140). If not in inventory, use equivalent locally purchased item. Reinstall the PC board.

- 8.42 Install fuse F2.

- 8.43 Complete maintenance procedures in Section 8.3.

8.5 Every 12 Years

- 8.51 Disconnect AC power, remove fuse F2, 2-1/2 amp. Replace the PC board and batteries. Perform maintenance procedures in section 8.20 YEARLY TEST.

- 8.52 Connect locally-procured 12 ohm, 5% or better, 1/4 watt or greater resistor between terminals 1 and 2 of TB2. Connect locally-procured 32 ohm, 5% or better, 1/4 watt or greater resistor between terminals 3 and 4 of TB2. Ensure that heater plug P2 is disconnected from J2. Place the TEST HEATER switch S2 in the A position and adjust R8 until the HEATER TEST LED just comes on. Place the TEST HEATER switch S2 in the B position and adjust R20 until the HEATER TEST LED just comes on. Remove the 12 and 32 ohm resistors.

8.53 Disconnect AC power, remove fuse F2 2-1/2 amp. Connect a variable 0 - 25 VDC 1 Amp external power supply to the test point labeled TP +20.7V positive and TP ground for negative. Connect a DVM across these test points as well. Slowly vary the voltage between 15 and 20 volts while adjusting R26 with the BATTERY TEST switch depressed until the BATTERY TEST LED lights at 16.5 volts and above, and does not light below 16.5 volts.

8.54 Remove the test DC power supply, replace fuse F2, and connect AC power.

WARNING: BEFORE CONNECTING P2 TO J2, ASSURE THAT AC POWER HAS BEEN APPLIED TO THE MRU FOR AT LEAST 3 MINUTES AND THAT THE RUN DOWN BUTTON HAS NOT BEEN ACTUATED FOR AT LEAST 3 MINUTES. FAILURE TO HEED THIS WARNING MAY CAUSE A MAGNET QUENCH.

8.55 Plug P2 into J2. Perform weekly test in Section 8.1. Log and date test results and items replaced.